

Smart Obstacles Detection System

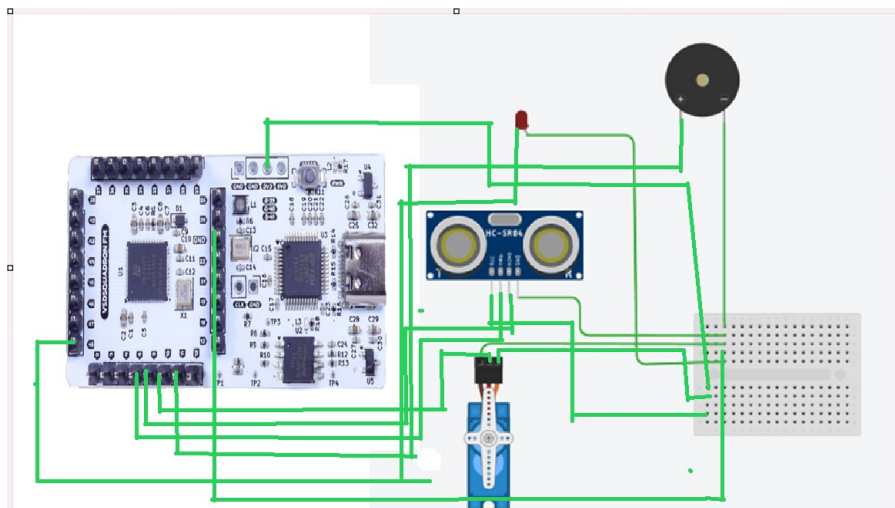
Overview

In this project we are using Lattice ICE40UP5K FPGA board .This project involves creating a system that detects obstacles using an ultrasonic sensor and alerts users with an LED and buzzer. A servo motor is used to rotate the ultrasonic sensor for surrounding scanning. The system can be programmed for sensitivity adjustments and has various applications in robotics, security, and industrial automation.

Components required

- Lattice ICE40UP5K FPGA board
- Ultrasonic Sensor
- Buzzer
- LED
- Bread Board
- Power supply
- Jumper Wires

PINOUT DIAGRAM



CIRCUIT CONNECTIONS

The circuit connections are as follows: The servo motor is connected to the ICE40UP5K microcontroller with its VCC pin linked to a 5V power supply and GND to ground. The SIGNAL pin is connected to pin 13 on the microcontroller, enabling PWM control for servo rotation. The ultrasonic sensor is also powered by 5V (VCC) and grounded (GND), with its TRIGG pin connected to pin 10 and ECHO pin to pin 11 on the microcontroller. This setup allows the microcontroller to trigger distance measurements and receive echo signals. The piezo electric buzzer's positive terminal is connected to

pin 9, and its GND to ground, enabling sound alerts. Similarly, the LED's positive terminal is connected to pin 8, and its GND to ground, allowing visual indications. These connections facilitate the system's functionality in obstacle detection and alerting.

PIN CONNECTIONS TABLE

ULTRASONIC SENSOR	ICE40UP5K
VCC	5V
GND	GND
TRIGG	10
ECHO	11
SERVO MOTOR	ICE40UP5K
VCC	5V
GND	GND
SIGNAL	13

PEIZO ELECTRIC BUZZER	ICE40UP5K
POSITIVE	9
GND	GND

LED	ICE40UP5K
POSITIVE	8
GND	GND